1.2.1 Show $r(n,k) = \sum_{l+m=n} r(l,i)\, r(m,j)$ when $i+j=k$.

Pf: $r(n,k) = \#\{v \in \mathbb{Z}^k : n = v_1^2 + \dots + v_k^2\}$.

$= \#\{(v,w) : v \in \mathbb{Z}^i, w \in \mathbb{Z}^j : n = v_1^2 + \dots + v_i^2 + w_1^2 + \dots + w_j^2\}$

$= \sum_{l+m=n} \#\{v \in \mathbb{Z}^i : l = v_1^2 + \dots + v_i^2\} \cdot \#\{w \in \mathbb{Z}^j : m = w_1^2 + \dots + w_j^2\}$

$= \sum_{l+m=n} r(l,i) \cdot r(m,j)$.

1.2.2 $N \in \mathbb{Z}_+$, $\gamma \in SL_2(\mathbb{Z}/N\mathbb{Z})$. Lift γ to $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{Z})$.

(a) 1) Show $\gcd(c, d, N) = 1$.

Pf: Lift means $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \gamma \ (N)$.

Thus $ad - bc \equiv 1 \ (N)$. $\Rightarrow$ $ad + (-b)c + tN = 1$.

$\Rightarrow \gcd(d, c, N) = 1$

2) Show $\gcd(c', d') = 1$ for some $\begin{cases} c' = c + sN \\ d' = d + tN \end{cases}$ $s, t \in \mathbb{Z}$.

Let $c \neq 0$. $g = (c, d)$. $N \neq 1$. $s = 0$.

Find $\begin{cases} t \equiv 1 \ (p) & \forall p \mid g. \\ \text{and } t \equiv 0 \ (p) & \forall p \nmid g, p \mid c \end{cases}$ by CRT. $\left(\bigoplus \frac{R}{(m_i)} \simeq R / \bigcap (m_i)\right)$. if $(m_i)+(m_j)=R$

Then if $q \mid (c, d')$ $\Rightarrow$ $q \mid c$, $q \mid d' = d + tN$.

$\Rightarrow$ $q \mid g$ and $q \mid tN$. $\overset{q \nmid N}{\Rightarrow}$ $q \mid g$. $q \mid t$. $q \nmid g$. no.

or $q \nmid g$ and $q \mid c$, $q \mid d + tN$. $\overset{q \nmid N}{\Rightarrow}$ $q \mid d$. $q \mid t$. $q \mid g$. no.

$\Rightarrow q = 1$.

1.2.2.(b) Show that some lift of σ $\begin{pmatrix} a+kN & b+lN \\ c' & d' \end{pmatrix} \in SL_2(\mathbb{Z})$.

Thus $SL_2(\mathbb{Z}) \to SL_2(\mathbb{Z}/N\mathbb{Z})$ surjects.

Pf: we want $(a+kN)d' - (b+lN)c' = 1$.

(i.e. $\begin{cases} a+kN = a' \\ b+lN = b' \end{cases}$ for some particular (a', b') has solution (k, l).

Find k, l s.t. $ad - bc \equiv 1 \ (N)$.

that is, $(a+kN)(d+tN) - (b+lN)c = 1$.

Thus pick (k, l) s.t. $kd + ta + kt = lc$.

$\Rightarrow k(d+t) - lc = -ta$.

1.2.3. (a) p prime, $e \in \mathbb{Z}_+$. Show $|SL_2(\mathbb{Z}/p^e\mathbb{Z})| = p^{3e}(1-\frac{1}{p^2})$.

Pf: Consider ef, $GL_2(\mathbb{Z}/p\mathbb{Z}) / SL_2(\mathbb{Z}/p\mathbb{Z}) \simeq (\mathbb{Z}/p\mathbb{Z})^\times$.

$|(\mathbb{Z}/p\mathbb{Z})^\times| = p-1$. $|GL_2(\mathbb{Z}/p\mathbb{Z})| = \#$ basis of $(\mathbb{Z}/p\mathbb{Z})^2$,

which is $\#(v_1, v_2) \in (\mathbb{Z}/p\mathbb{Z})^2$ that spans $(\mathbb{Z}/p\mathbb{Z})^2$.

which is $\underset{v_1 \neq 0}{(p^2-1)} \underset{v_2 \neq kv_1}{(p^2-p)}$.

Thus $|SL_2(\mathbb{Z}/p\mathbb{Z})| = p(p^2-1) = p^3(1-\frac{1}{p^2})$.

Now do induction. Since $SL_2(\mathbb{Z}) = \langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \rangle$,

those two matrices also generate $SL_2(\mathbb{Z}/p^e\mathbb{Z})$

thus $SL_2(\mathbb{Z}) \twoheadrightarrow SL_2(\mathbb{Z}/p^e\mathbb{Z})$.

$\forall A \in SL_2(\mathbb{Z}/p^e\mathbb{Z})$ ~~$M \equiv A \ (p^e)$~~ $B \equiv M \ (p^{e+1})$ $B \in SL_2(\mathbb{Z}/p^{e+1}\mathbb{Z})$

$M = A$, $M \in SL_2(\mathbb{Z})$.

Thus $\pi: SL_2(\mathbb{Z}/p^{e+1}\mathbb{Z}) \twoheadrightarrow SL_2(\mathbb{Z}/p^e\mathbb{Z})$

$I + p^e X \mapsto I$ $X \in M_2(\mathbb{Z}/p\mathbb{Z})$.

$\det(I + p^e X) = 1 + p^e \operatorname{tr} X = 1 \Rightarrow \operatorname{tr} X = 0. \Rightarrow |\ker \pi| = p^3$.

(1.2.37)

(b) Show $|SL_2(\mathbb{Z}/N\mathbb{Z})| = N^3 \prod_{p|N} (1-\frac{1}{p^2})$.

so this is also $[SL_2(\mathbb{Z}) : \Gamma(N)]$.

Pf: From (a) we see $|SL_2(\mathbb{Z}/p\mathbb{Z})| = p^3(1-\frac{1}{p^2})$.

Similarly $\pi_1 : SL_2(\mathbb{Z}/p_1^{e_1}p_2^{e_2}\mathbb{Z}) \twoheadrightarrow SL_2(\mathbb{Z}/p_1^{e_1}\mathbb{Z})$

$|\ker \pi_1| = p_2^{3e_2}$. $\quad I + p_1^{e_1}X \mapsto I \quad X \in M_2(\mathbb{Z}/p_2^{e_2}\mathbb{Z})$, $\operatorname{tr} X = 0$.

Assume $N = p_1^{e_1} \cdots p_n^{e_n}$.

then $|SL_2(\mathbb{Z}/N\mathbb{Z})| = \prod_{p_i|N} p_i^{3e_i}(1-\frac{1}{p_i^2}) = N^3 \prod_{p|N}(1-\frac{1}{p^2})$.

Since $SL_2(\mathbb{Z})/\Gamma(N) \xrightarrow{\sim} SL_2(\mathbb{Z}/N\mathbb{Z})$

$[SL_2(\mathbb{Z}) : \Gamma(N)] = N^3 \prod_{p|N}(1-\frac{1}{p^2})$.

(c). Show $\Gamma_1(N) \twoheadrightarrow \mathbb{Z}/N\mathbb{Z}$ and $\Gamma_1(N)/\Gamma(N) \xrightarrow{\sim} \mathbb{Z}/N\mathbb{Z}$.

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto b \ (N)$

Pf: $\Gamma_1(N) \hookrightarrow SL_2(\mathbb{Z}/N\mathbb{Z}) \twoheadrightarrow \mathbb{Z}/N\mathbb{Z}$ $\quad \varphi : \Gamma_1(N) \to \mathbb{Z}/N\mathbb{Z}$.

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}_{(N)} \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix}_{(N)} \mapsto b \ (N)$. $\quad \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}_{(N)} \mapsto b \ (N)$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{(N)} \mapsto 0 \ (N)$. so $\ker \varphi = \Gamma(N)$.

(d) Show $\Gamma_0(N) \twoheadrightarrow (\mathbb{Z}/N\mathbb{Z})^\times$. and

$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto d \ (N)$.

Pf: $\psi : \Gamma_0(N) \to (\mathbb{Z}/N\mathbb{Z})^\times$ $\quad \ker \psi = \Gamma_1(N)$.

$\forall d(N) \quad \psi^{-1}(d(N)) \ni \begin{pmatrix} d^{-1} & * \\ 0 & d \end{pmatrix}$. $d^{-1} \in (\mathbb{Z}/N\mathbb{Z})^\times$.

(e). Show $[SL_2(\mathbb{Z}) : \Gamma_0(N)] = N \prod_{p|N}(1+\frac{1}{p})$.

Pf: $[\Gamma_1(N) : \Gamma(N)] = \cancel{N-1}\ N$, so we need $|(\mathbb{Z}/N\mathbb{Z})^\times| = \varphi(N) = N\prod_{p|N}(1-\frac{1}{p})$

For p^e, $\varphi(p^e) = p^e - \underbrace{p^{e-1}}_{\text{(number of coprime)}} = p^e(1-\frac{1}{p})$. Thus $\varphi(N) = \prod_{p|N} p^e(1-\frac{1}{p}) = N\prod_{p|N}(1-\frac{1}{p})$

1.2.5. If Γ a congruence subgroup of $SL_2(\mathbb{Z})$. $\gamma \in SL_2(\mathbb{Z})$,

Then $\gamma^{-1}\Gamma\gamma$ again a congruence subgroup of $SL_2(\mathbb{Z})$.

Pf: $\exists N$ $\Gamma(N) \subset \Gamma \leq SL_2(\mathbb{Z})$. i.e. $\forall X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}(N) \in \Gamma$.

Then ~~γ^{-1}~~ $\gamma^{-1}X\gamma = \gamma^{-1}\left(I + \begin{pmatrix} k_1N & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & k_2N \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ k_3N & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & k_4N \end{pmatrix}\right)\gamma$.

$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$\gamma' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ $= I(N)$ ~~$a'k_1N$ $a'k_2N$~~

$X \mapsto X(N)$. gives $\Gamma(N) \mapsto \{I\}$

Actually recall $SL_2(\mathbb{Z}) \twoheadrightarrow SL_2(\mathbb{Z}/N\mathbb{Z})$, $SL_2(\mathbb{Z})/\Gamma(N) \simeq SL_2(\mathbb{Z}/N\mathbb{Z})$

Thus $\Gamma(N) \triangleleft SL_2(\mathbb{Z})$. i.e. $\gamma^{-1}\Gamma(N)\gamma = \Gamma(N)$.

Hence $\Gamma(N) = \gamma^{-1}\Gamma(N)\gamma \subset \gamma^{-1}\Gamma\gamma$, is congruence group of level N.

1.2.8. (a) Verify the Fourier expansion of G_2

Pf: $$G_2(\tau) = \sum_{(c,d)}{}' \frac{1}{(c\tau+d)^2} = \sum_{c\in\mathbb{Z}} \sum_{d\in\mathbb{Z}'_c} \frac{1}{(c\tau+d)^2}$$

$$\mathbb{Z}'_c = \begin{cases} \mathbb{Z}-\{0\} & c=0 \\ \mathbb{Z} & \text{otherwise} \end{cases}$$

From (1.2), $$\boxed{G_k(\tau) = 2\zeta(k) + 2\frac{(2\pi i)^k}{(k-1)!}\sum_{n=1}^{\infty}\sigma_{k-1}(n)q^n \quad k>2 \text{ even}}$$

$$\sum_{d\in\mathbb{Z}} \frac{1}{(\tau+d)^k} = \frac{(-2\pi i)^k}{(k-1)!}\sum_{m=1}^{\infty} m^{k-1}q^m \quad k\geq 2$$

$$q = e^{2\pi i\tau}$$

thus $$\sum_{d\in\mathbb{Z}} \frac{1}{(\tau+d)^2} = -4\pi^2\sum_{m=1}^{\infty} mq^m$$

$$G_2(\tau) = \sum_{d\in\mathbb{Z}-\{0\}} \frac{1}{d^2} + \sum_{c\in\mathbb{Z}-\{0\}}\sum_{d\in\mathbb{Z}} \frac{1}{(c\tau+d)^2}$$

$$= 2\zeta(2) + 2\sum_{c=1}^{\infty}(-4\pi^2)\sum_{m=1}^{\infty} mq^{cm}$$

$$= 2\zeta(2) - 8\pi^2\sum_{c=1}^{\infty}\sum_{m=1}^{\infty} mq^{cm} = 2\zeta(2) - 8\pi^2\sum_{n=1}^{\infty}\sigma(n)q^n, \quad \begin{pmatrix} d=m \\ n=cm \end{pmatrix}.$$

(b) i) assume G_2 satisfies (1.4)

$$(G_2[\gamma]_2)(\tau) = G_2(\tau) - \frac{2\pi i c}{c\tau+d} \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}).$$

for $\gamma_1, \gamma_2 \in SL_2(\mathbb{Z})$. show that

G_2 then satisfies for $\gamma_1\gamma_2$ ~~and γ_1~~

Pf: $(G_2[\gamma_1\gamma_2]_2)(\tau) =$ ~~$j(\gamma_1\gamma_2,\tau)^{-2} f(\gamma_1\gamma_2(\tau))$~~

Let $\gamma_1\gamma_2 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$= (G_2[\gamma_2]_2)(\tau) - \left(\frac{2\pi i c_1}{c_1\tau+d_1}[\gamma_2]_2\right)(\tau)$$

$$= G_2(\tau) - \frac{2\pi i c_1}{c_1\tau+d_1}\frac{1}{(c_2\tau+d_2)} - \frac{2\pi i c_2}{c_2\tau+d_2}$$

$$= G_2(\tau) - \frac{2\pi i c_2}{c_2\tau+d_2} - \frac{2\pi i c_1}{(c_2\tau+d_2)}\cdot\frac{1}{c_1\tau+d_1}$$

$$= G_2(\tau) - 2\pi i\frac{c_2(c\tau+d) + c_1}{(c_2\tau+d_2)(c\tau+d)}$$

$$= G_2(\tau) - 2\pi i\frac{c(c_2\tau+d_2)}{(c_2\tau+d_2)(c\tau+d)} = G_2(\tau) - 2\pi i\frac{c}{c\tau+d}.$$

1.2.11 $\mathcal{H}$ is a $GL_2^+(\mathbb{Q})$ module since $\operatorname{Im}(\gamma(\tau)) = \det\gamma \dfrac{\operatorname{Im}\tau}{|c\tau+d|^2}$.

Extend $j(\gamma,\tau) = c\tau+d$ to $\gamma \in GL_2^+(\mathbb{Q})$.

Extend $[\ ]_k$ to $GL_2^+(\mathbb{Q})$ by $[\ ]_k : \mathbb{C}^{\mathcal{H}} \times GL_2^+(\mathbb{Q}) \to \mathbb{C}^{\mathcal{H}}$, $(f,\gamma) \mapsto (\tau \mapsto \cdots)$

$$(f[\gamma]_k)(\tau) = (\det\gamma)^{k-1} j(\gamma,\tau)^{-k} f(\gamma(\tau)).$$

(a) Show $[\gamma\gamma']_k = [\gamma]_k[\gamma']_k \quad \forall \gamma,\gamma' \in GL_2^+(\mathbb{Q})$. (Lem 1.2.2 (c))

Pf: $(f[\gamma\gamma']_k)(\tau) = (\det\gamma\gamma')^{k-1} j(\gamma\gamma',\tau)^{-k} f(\gamma\gamma'(\tau))$

$$= (\det\gamma)^{k-1}(\det\gamma')^{k-1}\big((ca'+dc')\tau+(cb'+dd')\big)^{-k} f(\gamma(\gamma'(\tau))).$$

$$(f[\gamma]_k[\gamma']_k)(\tau) = (\det\gamma')^{k-1}(c'\tau+d')^{-k}(f[\gamma]_k)(\gamma'(\tau))$$

$$= (\det\gamma')^{k-1}(c'\tau+d')^{-k}(\det\gamma)^{k-1} j(\gamma,\gamma'(\tau))^{-k} f(\gamma(\gamma'(\tau))).$$

$$j(\gamma,\gamma'(\tau))^{-1} = c\,\frac{a'\tau+b'}{c'\tau+d'}+d$$

$$j(\gamma,\gamma'(\tau))(c'\tau+d') = c(a'\tau+b')+d(c'\tau+d')$$

$$= (ca'+dc')\tau+(cb'+dd')$$

$$= j(\gamma\gamma',\tau).$$

Thus $(f[\gamma\gamma']_k)(\tau) = (f[\gamma]_k[\gamma']_k)(\tau) \quad \forall \tau \in \mathcal{H}$.

(b) 1) Show $\forall \gamma \in GL_2^+(\mathbb{Q}) \ \exists \alpha \in SL_2(\mathbb{Z})$ $\gamma = \alpha\gamma'$

where $\gamma' = r\begin{pmatrix} a & b \\ 0 & d\end{pmatrix}$ $r \in \mathbb{Q}^+$ $a,b,d \in \mathbb{Z}$ relatively prime

Pf: Let $\gamma = \begin{pmatrix} a & b \\ c & d\end{pmatrix}$. If $c=0$, then $r = \dfrac{\prod \text{denominator of } a,b,d}{\gcd \text{ factor of } a,b,d}$, $\alpha = I$.

If $c \neq 0$, then $\gamma = \gcd(a,c)\begin{pmatrix} a' & b' \\ c' & d'\end{pmatrix}$, $c' = c\cdot\dfrac{a'}{a}$, so that $\dfrac{a'}{c'} = \dfrac{a}{c}$ reduced.

Let $\alpha_1 = \begin{pmatrix} * & * \\ c' & -a'\end{pmatrix}$, $\alpha_1\begin{pmatrix} a' & b' \\ c' & d'\end{pmatrix} = \begin{pmatrix} * & * \\ 0 & *\end{pmatrix}$ turns into case 1.

((1.2.11) 2) Use this to show $\forall f \in M_k(\Gamma)$, $f[\gamma]_k$ has Fourier expansion, with ~~some~~ constant term 0 if $f[\alpha]_k$ has $a_0 = 0$.

if: $f[\gamma]_k = f[\alpha\gamma']_k = f[\alpha]_k \left[r\begin{pmatrix} a & b \\ 0 & d\end{pmatrix}\right]_k$.

First write $f[\alpha]_k$ has period h, then

$$(f[\alpha]_k)(\tau) = \sum_{n\geq 0} a_n e^{\frac{2\pi i n\tau}{h}} =: \sum_{n\geq 0} a_n q^n.$$

Now apply γ', $\left(f[\alpha]_k \left[r\begin{pmatrix} a & b \\ 0 & d\end{pmatrix}\right]_k\right)(\tau)$

$$= (rd)^{-k}(f[\alpha]_k)\left(\frac{ra\tau+rb}{rd}\right) = (rd)^{-k}(f[\alpha]_k)\left(\frac{a\tau+b}{d}\right)$$

$$= (rd)^{-k}\sum_{n\geq 0} a_n e^{\frac{2\pi i n(a\tau+b)}{dh}} = (rd)^{-k}\sum_{n\geq 0}\left(a_n e^{\frac{2\pi i n b}{dh}}\right)\cdot e^{\frac{2\pi i n\tau}{\frac{dh}{a}}}$$

Thus $f[\gamma]_k$ is of period $\frac{dh}{a}$.

new constant term is $(rd)^{-k} a_0 e^{\frac{2\pi i 0\cdot b}{dh}} = (rd)^{-k} a_0$.

(c) 1) Suppose Γ_1, Γ_2 are congruence subgroups.

suppose $\Gamma_1 \supset \gamma\Gamma_2\gamma^{-1}$, $\gamma \in GL_2^+(\mathbb{Q})$.

Prove if $f \in M_k(\Gamma_1)$ then $f[\gamma]_k \in M_k(\Gamma_2)$.

Pf: $f \in M_k(\Gamma_1) \Rightarrow (f[\gamma_1]_k)(\tau) = j(\gamma_1,\tau)^{-k} f(\gamma_1(\tau)) = f(\tau)$.

Then $(f[\gamma_1]_k[\gamma_2]_k)(\tau) = (f[\gamma_1\gamma_2]_k)(\tau) = (f[\gamma_2\gamma_1]_k)(\tau)$.

$= (f[\gamma_2]_k[\gamma_1]_k)(\tau) = (f[\gamma]_k)(\tau)$.

2) Show the same for cusp forms.

Pf: If $f \in S_k(\Gamma_1)$, then $(f[\alpha]_k)(\tau) = \sum_{n>0} a_n q^n$ $\forall \alpha \in SL_2(\mathbb{Z})$.

Then $(f[\gamma_1]_k[\alpha]_k)(\tau) = (f[\alpha]_k[\gamma_1]_k)(\tau)$

$= (rd)^{-k}(f[\alpha]_k[\alpha']_k)\left(\frac{a\tau+b}{d}\right) = (rd)^{-k}(f[\alpha\alpha']_k)\left(\frac{a\tau+b}{d}\right)$.

$\alpha\alpha' \in SL_2(\mathbb{Z})$. Thus $(f[\gamma]_k[\alpha]_k)(\tau) = (rd)^{-k}\sum_{n>0} a_n e^{\frac{2\pi i b n}{dh}}\cdot e^{\frac{2\pi i\tau}{\frac{dh}{a}}}$

is a cusp form.

(1.2.11)(d) 1) Let $N, d \in \mathbb{Z}_+$. $f \in M_k(\Gamma_0(N))$.

Show if $g(\tau) = f(d\tau)$ then $g \in M_k(\Gamma_0(dN))$ and the same for cusp forms.

Pf: $f \in M_k(\Gamma_0(N)) \Rightarrow (f[\gamma]_k)(\tau) = f(\tau)$

$\forall \gamma \equiv \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \mod N$.

Let $\gamma' \equiv \gamma \mod d$, $\gamma' = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}$

$\Rightarrow (g[\gamma']_k)(\tau) = \cancel{f(\gamma)}\ d'^{-k} g(\gamma'(\tau))$

$= d'^{-k} g(d\gamma'(\tau))$.

Set $\Gamma_1 = \Gamma_0(N)$, $\Gamma_2 = \Gamma_0(dN)$. To find $\gamma \in GL_2^+(\mathbb{Q})$.

Let $\gamma = \begin{pmatrix} d & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \gamma^{-1} = \begin{pmatrix} \frac{1}{d} & 0 \\ 0 & 1 \end{pmatrix}$.

$\forall M = \begin{pmatrix} a & b \\ c & d' \end{pmatrix} \in \Gamma_0(dN)$, $\gamma M \gamma^{-1} = \underbrace{\begin{pmatrix} a & bd \\ c/d & d' \end{pmatrix}}_{=: M'} \in \Gamma_0(N)$.

Thus (1) tells the result.

$\cancel{(g[M]_k)(\tau) =}$ $(f[\gamma]_k)(\tau) = f(d\tau)$.

2) Show if $f \in M_k(\Gamma_1(N))$ then $g \in M_k(\Gamma_1(dN))$ and the same for cusp forms.

Pf: Set $\Gamma_1' = \Gamma_1(N)$, $\Gamma_2' = \Gamma_1(dN)$. $\forall M \in \Gamma_2'$

$M' = \gamma M \gamma^{-1}$ in 1) also lives in Γ_1'. Thus holds as well.